



Dr. Walaa Al Shuaibi
King Khalid University Hospital
Pediatrics

Riyadh
Saudi Arabia

Order no.: 62451890

Order received: 09 Mar. 2018

Sample type: blood, filter card

Sample collection date: 18 Feb. 2018

Report date: 12 Apr. 2018

Report type: Final Report



Patient no.: **1223557**, First Name: **Meshael**, Last Name: **Alsubaie**
DOB: **19 Dec. 2011**, Sex: **male**, Your ref.: **01-20-89-66**

Test(s) requested: Whole Exome Sequencing (CentoXome GOLD®)

CLINICAL INFORMATION*

Chewing difficulties, Delayed gross motor development, Easy fatigability, Elevated hepatic transaminases, Elevated serum creatine phosphokinase, Global developmental delay, Muscle weakness, Skeletal muscle hypertrophy

*: Clinical information indicated above follows HPO nomenclature.

Age of manifestation: 18 months

Previous sequence analysis of our congenital myopathy panel genes resulted negative (report 62401032 from 21 Aug. 2017).

Please also see our additional report regarding chromosomal microarray analysis, which has been performed in parallel.



POSITIVE RESULT
Likely pathogenic variant identified

INTERPRETATION

A hemizygous likely pathogenic nonsense variant was identified in the DMD gene. **The genetic diagnosis of Duchenne Muscular Dystrophy is thus confirmed.**

Of note, patients with nonsense mutations, such as the one detected here, can benefit from novel genetic therapies implementing stop codon read-through technology.

RECOMMENDATIONS

- maternal carrier testing to evaluate whether the detected variant is inherited or *de novo*
- genetic counselling

> Contact Details

Tel.: +49 (0)381 80113 416
Fax: +49 (0)381 80113 401
dmqc@centogene.com
www.centogene.com

CLIA registration 99D2049715; CAP registration 8005167. Scientific use of these results requires permission of CENTOGENE. If you would like to download your reports from our web portal, please contact us to receive your login and password. More information is available at www.centogene.com or support@centogene.com.





RESULT SUMMARY

GENE	VARIANT COORDINATES	ZYGOSITY	IN SILICO PARAMETERS*	ALLELE FREQUENCIES**	TYPE AND CLASSIFICATION***
DMD	ChrX(GRCh37):g.32503197G>C NM_004006.2:c.2642C>G p.(Ser881*) Exon 21	Hem	PolyPhen: N/A Align-GVGD: N/A SIFT: N/A MutationTaster: N/A Conservation: nt moderate	gnomAD: - ESP: - 1000 G: - CentoMD: -	Stop gain Likely pathogenic (class 2)

Variant description based on Alamut Batch (latest database available). * AlignGVGD: C0: least likely to interfere with function, C65: most likely to interfere with function, splice prediction tools: SSF, MaxEnt, HSF. ** Genome Aggregation Database (gnomAD), Exome Sequencing Project (ESP), 1000Genome project (1000G) and CentoMD® (latest database available). *** based on ACMG recommendations

VARIANT INTERPRETATION

DMD, c.2642C>G p.(Ser881*)

The DMD variant c.2642C>G p.(Ser881*) creates a premature stop codon. It is classified as likely pathogenic (class 2) according to the recommendations of Centogene and ACMG (please, see additional information below).

Pathogenic variants in DMD gene are associated with Duchenne muscular dystrophy, an X-linked disorder. X-linked recessive dystrophin-associated muscular dystrophies range from the severe Duchenne muscular dystrophy (DMD, OMIM®: 310200) to the milder Becker muscular dystrophy (BMD; OMIM®: 300376). DMD is featured by a progressive proximal muscular dystrophy with characteristic pseudohypertrophy of the calves. The bulbar (extraocular) muscles are spared but the myocardium is affected. There is massive elevation of creatine kinase levels in the blood, myopathic changes by electromyography, and myofiber degeneration with fibrosis and fatty infiltration on muscle biopsy. The onset of DMD usually occurs before age 3 years, and the patient is chair ridden by age 12. The onset of BMD is often in the 20s and 30s and survival to a relatively advanced age is frequent.

INCIDENTAL FINDINGS

Incidental findings which we list according to the ACMG guidelines are not provided here due to the lack of consent.

ANALYSIS STATISTICS WES

AVERAGE COVERAGE (X)	% TARGET BP COVERED					
	0X	≥ 1X	≥ 5X	≥ 10X	≥ 20X	≥ 50X
96.94	0.13	99.87	99.35	98.13	93.51	69.45

CENTOGENE VARIANT CLASSIFICATION (BASED ON ACMG RECOMMENDATIONS)

Class 1 – Pathogenic

Class 2 – Likely pathogenic

Class 3 – Variant of uncertain significance (VUS)

Additionally, other types of clinical relevant variants can be identified (e.g. risk factors, modifiers).

Class 4 – Likely benign

Class 5 – Benign

METHODS

RNA capture baits against approximately 60 Mb of the Human Exome (targeting >99% of regions in CCDS, RefSeq and Ensembl databases) is used to enrich regions of interest from fragmented genomic DNA with Agilent's SureSelect Human All Exon V6 kit. The generated library is sequenced on an Illumina platform to obtain an average coverage depth of ~100x. Typically, ~97% of the targeted bases are covered >10x. An end to end in-house bioinformatics pipeline including base calling, alignment of reads to GRCh37/hg19 genome assembly, primary filtering out of low quality reads and probable artefacts, and subsequent annotation of variants, is applied. All disease causing variants reported in HGMD®,

> Contact Details

Tel.: +49 (0)381 80113416
Fax: +49 (0)381 80113401
dmqc@centogene.com
www.centogene.com

CLIA registration 99D2049715; CAP registration 8005167. Scientific use of these results requires permission of CENTOGENE. If you would like to download your reports from our web portal, please contact us to receive your login and password. More information is available at www.centogene.com or support@centogene.com.





in ClinVar or in CentoMD® as well as all variants with minor allele frequency (MAF) of less than 1% in gnomAD database are considered. Evaluation is focused on coding exons along with flanking +/-20 intronic bases. All pertinent inheritance patterns are considered. In addition, provided family history and clinical information are used to evaluate eventually identified variants. All identified variants are evaluated with respect to their pathogenicity and causality, and these are categorized into classes 1 - 5 (see above). All variants related to the phenotype of the patient, except benign or likely benign variants, are reported. Variants of relevance identified by NGS are continuously and individually in-house validated for quality aspects; those variants which meet our internal QC criteria (based on extensive validation processes) are not validated by Sanger.

LIMITATIONS

Test results are interpreted in the context of clinical findings, family history and other laboratory data. Only variations in genes potentially related to the proband's medical condition are reported. Rare polymorphisms may lead to false negative or positive results. Misinterpretation of results may occur if the information provided is inaccurate or incomplete. If results obtained do not match the clinical findings, additional testing should be considered.

Specific genetic events like copy number variants, translocations and repeat expansions may not be reliably detected with Exome Sequencing. In addition, due to limitations in technology, certain regions may either not be covered or may be poorly covered, where variants cannot be confidently detected.

ADDITIONAL INFORMATION

This test was developed and its performance validated by CENTOGENE AG. The US Food and Drug Administration (FDA) has determined that clearance or approval of this method is not necessary and thus neither have been obtained. This test has been developed for clinical purposes. All test results are reviewed, interpreted and reported by our scientific and medical experts.

In line with ACMG recommendations for reporting of incidental findings in clinical exome and genome sequencing (Genetics in Medicine, 2016), we report incidental findings, i.e. pathogenic variants (class 1) and likely pathogenic variants (class 2) only in the recommended genes for the recommended phenotypes.

To also exclude mistaken identity in your clinic, several guidelines recommend testing a second sample that is independently obtained from the proband. Please note that any further analysis will result in additional costs.

The classification of variants can change over the time. Please feel free to contact CENTOGENE (dmqc@centogene.com) in the future to determine if there have been any changes in classification of any reported variants.

DISCLAIMER

Any preparation and processing of a sample from patient material provided to CENTOGENE by a physician, clinical institute or a laboratory (by a "Partner") and the requested genetic and/or biochemical testing itself is based on the highest and most current scientific and analytical standards. However, in very few cases genetic or biochemical tests may not show the correct result, e.g. because of the quality of the material provided by a Partner to CENTOGENE or in cases where any test provided by CENTOGENE fails for unforeseeable or unknown reasons that cannot be influenced by CENTOGENE in advance. In such cases, CENTOGENE shall not be responsible and/or liable for the incomplete, potentially misleading or even wrong result of any testing if such issue could not be recognized by CENTOGENE in advance.

COPYRIGHT NOTICE

This document contains information from the Online Mendelian Inheritance in Man® (OMIM®) database, which has been obtained under a license from the Johns Hopkins University. This document does not represent the entire, unmodified OMIM® database, which is available in its entirety at <http://omim.org/downloads>. Regarding OMIM® information: Copyright © 1996 – 2017, John Hopkins University, all rights reserved.

Prof. Dr. Arndt Rolfs, MD
Chief Medical Director

Dr. Iris Hövel, PhD
Clinical Scientist

> Contact Details

Tel.: +49 (0)381 80113416
Fax: +49 (0)381 80113401
dmqc@centogene.com
www.centogene.com

CLIA registration 99D2049715; CAP registration 8005167. Scientific use of these results requires permission of CENTOGENE. If you would like to download your reports from our web portal, please contact us to receive your login and password. More information is available at www.centogene.com or support@centogene.com.

